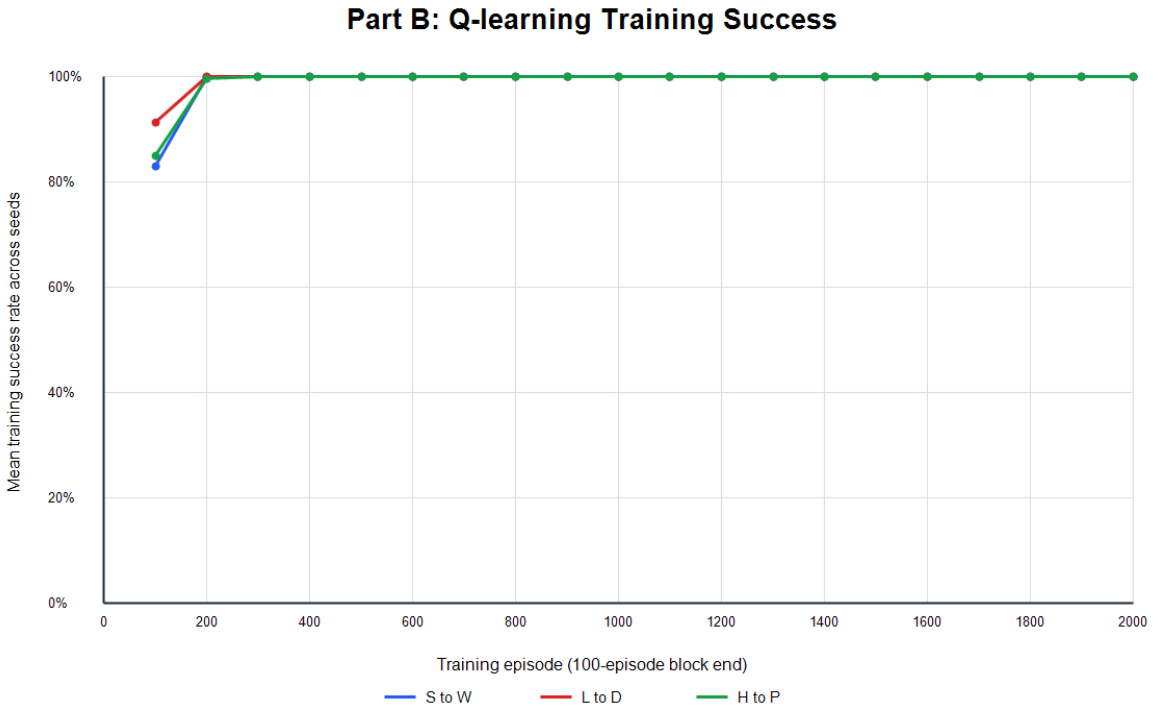
Part A

The search problem's state is the robot's traversable grid position, and the initial state is the chosen starting building entrance. The available actions are up, down, left, and right and the known map determines the transition model: a valid move leads to the adjacent traversable cell, while a blocked move leaves the robot in place. The goal test checks whether the robot has reached the destination entrance, and the path cost is the number of moves, with each move costing 1. Because every move has equal cost, BFS explores states in order of increasing path length and therefore finds an optimal, shortest route.

Map	**Delivery Task**	**BFS Route Length**	**A* Route Length**	**BFS Expanded Nodes**	**A* Expanded Nodes**
Baseline	S->W	18	18	67	22
Baseline	L->D	22	22	79	62
Baseline	H->P	16	16	41	16
Detour	S->W	26	26	70	51
Detour	L->D	22	22	65	60
Detour	H->P	16	16	41	16

Part B

In Part B, the Q-learning agent learned effective routes for all three delivery tasks. During evaluation, each task achieved a 100% success rate across all three seeds, with average delivery times of 18 steps for S -> W, 22 for L -> D, and 16 for H -> P. Training success was already fairly high in the first 100 episodes and reached 100% by episode 200 for S -> W and L -> D, and by episode 300 for H -> P. These results suggest the agent learned the routes quickly under the tested settings, with no movement slip. Since the evaluation results were identical across the three seeds, performance was consistent in these runs.



Part C

(i) My question was: How does the exploration rate affect Q-learning on my custom map's S -> W delivery task? I predicted that higher exploration might slow early learning, but that all three settings could eventually learn a successful route.

(ii) My custom map is an 11-by-15 grid with a blocked border and all six building entrances. The open walkways contain a loop in the upper-left area and a dead-end branch at row 8, column 4 (using one-based grid positions). S -> W is reachable, and the shortest route is 16 steps.

(iii) I trained the agent on the custom map's S -> W task with exploration rate values of 0.05, 0.2, and 0.5, using seeds 0, 1, and 2 for each setting. I kept the other parameters at their default 2,000 training episodes, a = 0.3, y = 0.95, no slip, 100 evaluation episodes per seed, and a 200-action limit and evaluated with a greedy policy and learning frozen.

(iv) All three exploration rate settings achieved 100% evaluation success across the seeds, with an average delivery time of 16 steps and a mean return of 24. The first 100-episode training block averaged 90.67% success for exploration rate = 0.05, 86.33% for exploration rate = 0.2, and 89.00% for exploration rate = 0.5; in the 101–200 block, all three averaged 100%.

Exploration Rate	Training Success,	Training Success,	Evaluation Success	Mean Return	Mean Delivery

	Episodes 1-100	episodes 101-200	Rate		Steps
0.05	90.67%	100%	100%	24	16
0.2	86.33%	100%	100%	24	16
0.5	89.00%	100%	100%	24	16

Map	Delivery Task	BFS route length	A* route length	BFS expanded nodes	A* expanded nodes
my_campus.txt	S -> W	16	16	93	36

(v) The results suggest that, for this map and these training settings, changing exploration rate affected early training success somewhat but did not change final evaluation performance. This was consistent with my prediction that all settings could eventually learn a successful route, although the differences in early success were small and did not show a clear trend as exploration rate increased.